

## SUMMARY TABLE: GENERAL SCIENCE GRADE 10

### Sub-Strand 3.2: Linear Motion — Teacher Reference (Pre-filled)

| SUMMARY TABLE: GENERAL SCIENCE GRADE 10 |                                                                                                                                                                                                        |
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| Sub-Strand                              | Sub-Strand 3.2: Linear Motion                                                                                                                                                                          |
| Driving Question                        | DRIVING QUESTION: Why does a runaway mkokoteni on the Limuru escarpment keep getting faster — and how can we use mathematics to predict exactly how fast and how far it will travel before it crashes? |

### INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

| Lesson # and Title                                                                                                  | What did I observe?                                                                                                                                                                                                                                                         | What did I learn?                                                                                                                                                                                                                                                                                                                                                                                     | How does this explain the phenomenon?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>Lesson 1</b><br><b>Anchoring Phenomenon &amp; Motion Vocabulary: Why Does the Mkokoteni Keep Getting Faster?</b> | A video clip of a runaway vehicle accelerating freely down a steep Kenyan escarpment road, showing continuous increase in speed with no engine or external push; sticky-note puzzles generated by student groups about the cause of the increasing speed.                   | The five key terms of linear motion — distance, displacement, speed, velocity, and acceleration — and their differences, anchored in the mkokoteni scenario and Kenyan athletics examples (Faith Kipyegon, Nyayo Stadium, Nairobi–Mombasa matatu). Students learned that the mkokoteni's continuous speed increase is called acceleration, and that gravity is the likely cause.                      | Students cannot yet explain WHY gravity causes a constant, predictable acceleration, nor can they calculate how fast or how far the mkokoteni will travel at any given time. These gaps — captured in the DQB and in the Initial Model sentence starters — form the questions that Lessons 2–6 will answer through equations of linear motion and the concept of free fall.                                                                                                                                                                                                             |
| <b>Lesson 2</b><br><b>Equations of Linear Motion (Part 1): Giving the Mkokoteni a Formula</b>                       | The mkokoteni started from rest on the Limuru escarpment and accelerated continuously. Witnesses said it became dangerously fast within seconds. We knew from Lesson 1 that acceleration means a constant change in velocity, but we could not yet put exact numbers to it. | There are four equations of linear motion that apply whenever acceleration is constant. The five variables are $u$ (initial velocity), $v$ (final velocity), $a$ (acceleration), $s$ (displacement), and $t$ (time). We select the equation that matches the three variables we know and the one we want to find. The mkokoteni starting from rest means $u = 0$ , which simplifies the calculations. | Using $v = u + at$ we calculated that a mkokoteni starting from rest with an acceleration of $2 \text{ m/s}^2$ reaches $12 \text{ m/s}$ after just 6 seconds — fast enough to be lethal. Using $s = ut + \frac{1}{2}at^2$ we found it had already rolled 36 m by that point. This gives us the first mathematical evidence toward answering the driving question: we CAN predict how fast and how far the cart travels, as long as we know the acceleration. The remaining puzzle is: what is the exact value of gravitational acceleration — and we will investigate that in Lesson 4. |

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| <b>Lesson 3</b><br><b>Equations of Linear Motion — Part 2: Problem Solving</b>                                                                | <p>Students observed three worked problem contexts — a Rift Valley athlete accelerating, a boda-boda decelerating before a Nairobi police check-point, and a ball thrown upward from a Mombasa rooftop — and observed peer groups presenting full solutions on the board, including cases where deceleration was treated as negative acceleration and where vertical free-fall problems required careful sign conventions.</p> | <p>Students learned how to systematically identify GIVEN and FIND quantities before selecting an equation; that the four equations of linear motion can predict speed, distance, and time for any constant-acceleration scenario; that deceleration is negative acceleration and free-fall uses <math>g = 10 \text{ m/s}^2</math> downward; and that the runaway mkokoteni reaches approximately 20 m/s after only 8 seconds from rest, covering 80 m — making it lethally fast before it even reaches the base of the Limuru escarpment.</p>                                                                      | <p>Students explained why the mkokoteni keeps accelerating without anyone pushing it (gravity provides a continuous unbalanced force down the slope), and used the equations <math>v = u + at</math> and <math>s = ut + \frac{1}{2}at^2</math> to produce the first fully quantitative predictions for the cart's speed and distance — directly answering the driving question's demand to predict 'exactly how fast and how far' the cart travels before crashing.</p>                                                                      |
| <b>Lesson 4</b><br><b>Investigating Free Fall: Does Mass Matter? Dropping Objects from the Limuru Escarpment Height</b>                       | <p>We dropped a rubber bung, a crumpled paper ball and a small stone from exactly 2 m. In every trial where we dropped two objects simultaneously, they hit the ground at the same time — even though they had different masses. We timed individual drops and recorded the height and fall time.</p>                                                                                                                          | <p>Using <math>h = \frac{1}{2}gt^2</math>, we calculated <math>g</math> from our experimental data and found a class average close to <math>9.8 \text{ m/s}^2</math>. This confirms that gravity gives ALL falling objects the same acceleration regardless of mass. After 3 seconds of free fall (or free rolling with no friction), the mkokoteni would be travelling at approximately 29.4 m/s — a speed that explains why witnesses described it as dangerously fast within seconds.</p>                                                                                                                       | <p>Gravity exerts a downward force (weight, <math>W = mg</math>) on every object. Although a heavier mkokoteni experiences a greater gravitational force, it also has more mass to accelerate. The two effects cancel exactly (<math>a = F/m = mg/m = g</math>), so every object — light mango or heavy cart — accelerates downward at the same rate of <math>9.8 \text{ m/s}^2</math>. This constant acceleration is why the mkokoteni's speed increases by 9.8 m/s every second, even with no engine and no one pushing it.</p>            |
| <b>Lesson 5</b><br><b>Freely Falling Bodies — Explain &amp; Calculate: Using <math>g</math> to Predict the Mkokoteni's Speed and Distance</b> | <p>Students verified <math>g \approx 10 \text{ m/s}^2</math> using their own Lesson 4 drop data, confirming that the mkokoteni on the Limuru escarpment and a mango falling from a tree in Kisumu experience the same gravitational acceleration.</p>                                                                                                                                                                          | <p>When a body is released from rest (<math>u = 0</math>) under gravity (<math>a = g = 10 \text{ m/s}^2</math>), its motion is fully described by three simplified equations: <math>v = at</math> (speed after time <math>t</math>); <math>s = \frac{1}{2}at^2</math> (distance after time <math>t</math>); <math>v^2 = 2as</math> (speed after distance <math>s</math>). Applying these equations shows that the mkokoteni reaches approximately 50 m/s (180 km/h) after only 5 seconds — far beyond any safe road speed — explaining why witnesses described it as becoming dangerously fast within seconds.</p> | <p>Gravity is an unbalanced force that continuously accelerates any object released from rest on a slope or in free fall. Because the force never stops acting, speed increases every second by <math>g \approx 10 \text{ m/s}^2</math>, making the mkokoteni's acceleration inevitable without an engine. The three equations of linear motion transform this physical insight into precise, testable predictions — answering the driving question: we CAN predict exactly how fast and how far the cart will travel before it crashes.</p> |
| <b>Lesson 6</b><br><b>Applications, Road Safety &amp; Real-Life Contexts of Linear Motion</b>                                                 | <p>Students calculated braking distances for a matatu at 30 km/h, 60 km/h, and 90 km/h on Mombasa Road using <math>v^2 = u^2 + 2as</math>, graphed speed vs. braking distance (observing a non-linear/quadratic curve), and examined an NTSA evidence card showing a warning sign on the Kinangop</p>                                                                                                                          | <p>Braking distance is proportional to the square of initial speed (<math>s = u^2/2a</math> when <math>v = 0</math>), meaning that tripling speed increases stopping distance nine times — the same quadratic relationship that governs how fast the mkokoteni built up dangerous speed on the Limuru escarpment. NTSA road-safety</p>                                                                                                                                                                                                                                                                             | <p>Because braking distance grows as the square of speed (not linearly), small increases in vehicle speed produce disproportionately large increases in stopping distance — exactly as gravity produced a disproportionately dangerous speed increase in the runaway mkokoteni.</p>                                                                                                                                                                                                                                                          |

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|                                                                                                                | descent alongside a statistic that 60% of fatal crashes on Kenyan roads involve excessive speed on slopes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | guidelines and steep-road warning signs are direct policy applications of this physics. The three kinematic equations apply equally to the mkokoteni, a braking matatu, an SGR train, a falling construction object, and an athlete's run-up.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | The same three equations of linear motion ( $v = u + at$ ; $s = ut + \frac{1}{2}at^2$ ; $v^2 = u^2 + 2as$ ) that let us predict the cart's speed at any moment also let engineers calculate safe following distances, set speed limits on escarpment roads, schedule SGR trains, and design construction-site safety protocols — demonstrating that mathematics is a universal, life-saving tool.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Lesson 7</b><br><b>Group Presentations and Collaborative Model Building</b>                                 | Students wrote individual journal predictions estimating the largest acceleration among their classmates researched applications, citing numerical values and equations of motion. Each group then delivered a 3-minute presentation on a real-life linear motion application — including contexts such as SGR train scheduling, pole-vaulting, and construction falling-object safety — while the rest of the class recorded the application name, the equation or equations used, and the connection to the mkokoteni in a three-column listening frame. The teacher cold-called two members per group for follow-up questions, and students physically sorted the Driving Question Board cards from Lessons 1 to 6 into answered and still-investigating columns, annotating each card with the lesson that provided the evidence. | The class established that every application presented — whether an SGR train accelerating along a scheduled route, a pole-vaulter building speed during a run-up, or a concrete block falling on a construction site — can be fully described using the same five variables ( $u$ , $v$ , $a$ , $s$ , $t$ ) and the same four equations of linear motion studied earlier in this unit. Students consolidated the understanding that gravitational acceleration $g$ is the single unifying constant linking free fall, the mkokoteni on the escarpment, and all real-world falling or decelerating scenarios. A collaboratively built, wall-sized annotated model of the mkokoteni descending the Limuru escarpment brought all sub-strand concepts together in one place: distance markers, proportionally scaled velocity arrows labelled with calculated $v$ values, the symbol $g$ , braking-distance annotations, and safety mitigation arrows showing where intervention could prevent a crash. | Pooling evidence from all group presentations and from lessons studied earlier in this unit, the class could now show on a single visual model exactly why the runaway mkokoteni on the Limuru escarpment keeps getting faster: gravity provides a constant unresisted acceleration, so velocity grows continuously with every metre the cart travels downhill, precisely as the growing arrows on the model demonstrate. The annotated distance markers at 0 m, 50 m, 100 m, and 200 m gave concrete, calculated evidence that the cart reaches a dangerously high speed long before any obstacle, directly answering the first part of the driving question. The safety mitigation arrows added to the model linked back to the road-safety discussion from the lesson studied immediately before this one, showing that understanding the mathematics of constant acceleration is not merely academic — it is what makes it possible to predict, and therefore prevent, a crash like the one that opened the unit. |
| <b>Lesson 8</b><br><b>Consolidation, DQB Completion &amp; Final Explanation — The Mkokoteni Mystery Solved</b> | Students re-watched the mkokoteni video clip and individually completed a six-problem structured assessment covering all four equations of linear motion and free fall, set within the contexts of the Limuru escarpment cart, a Kenyan sprinter at Nyayo Stadium, and a falling mango in Kisumu.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Gravity provides a constant unbalanced force on the mkokoteni, producing constant acceleration ( $a \approx 3 \text{ m/s}^2$ ). This means speed increases linearly with time ( $v = u + at$ ) and distance increases as the square of time ( $s = ut + \frac{1}{2}at^2$ ). Using $v^2 = u^2 + 2as$ , students confirmed that the cart reached approximately 40 m/s (144 km/h) at the crash point — consistent with witness accounts. Free fall ( $g = 10 \text{ m/s}^2$ ) is the limiting case of this same gravitational acceleration when no slope component or friction is present. All four equations are                                                                                                                                                                                                                                                                                                                                                                                        | The mkokoteni kept accelerating because gravity — an invisible but constant force — continuously transferred energy to the cart throughout its descent, producing a uniform acceleration of approximately $3 \text{ m/s}^2$ . The equations of linear motion are not arbitrary formulas but precise mathematical descriptions of this physical reality, enabling anyone — engineer, driver, or student — to predict the cart's exact speed and position at every moment of its descent, and therefore to design evidence-based safety interventions such as crash barriers placed                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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|  |  | mathematically equivalent tools for predicting motion under constant acceleration. | at calculated distances along the Limuru–Nairobi escarpment road. |
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